

Yao Fu

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Research

I study the intersection of **machine learning** and **distributed systems**. My goal is to build efficient systems for the large-scale deployment of machine learning models. My current research focuses on the efficient inference of **large language models** in **serverless computing** clusters.

Education

The University of Edinburgh Ph. D. - Computer Science; Supervisor: Prof. Luo Mai	Edinburgh, UK Sept. 2021 - present
Sun Yat-sen University B. E. - Computer Science and Technology	Guangzhou, China Sept. 2017 - Jun. 2021

Selected Research Projects

ServerlessLLM : Locality-Enhanced Serverless Inference for Large Language Models	Apr. 2022 - Present
- Developed a loading-optimized checkpoint format and a fast checkpoint loader. (4X faster than SafeTensor) - Designed a live-migration mechanism for locality-driven LLM inference. (2X better than serverless scheduling policies) - Designed a model loading scheduler for locality-aware server allocation (reducing start-up latency by 1.86X). - Evaluated Phantom against SOTA serverless inference systems: Ray Serve and KServe, showing 10 - 200X speed up. - GitHub Project: ServerlessLLM/ServerlessLLM (https://github.com/ServerlessLLM/ServerlessLLM)	
MoE-Infinity : Activation-Aware Expert Offloading for Efficient MoE Serving	Apr. 2022 - Present
- Evaluated tensor prefetching and caching policies for Mixture of Experts model inference. - Developed Archer's Python binding, compatible with HuggingFace Transformers. - Evaluated Archer with DeepSpeed Infinity, showing 9X performance improvement. - GitHub Project: TorchMoE/MoE-Infinity (https://github.com/TorchMoE/MoE-Infinity)	

Work Experience

Tencent Research Intern, MLSys Team; Mentor: Feng Lin	Guangzhou, China May 2021 - Jan. 2022
- Designed an SLO-aware model update scheduler for a large-scale Deep Learning Recommender System - Proposed an inference model state manager to monitor model health and implement low-latency rollbacks. - Mitigated a 2.32% SLO drop during network congestion in real-world Short Video services with over one billion users.	

Selected Publications

*Co-primary authors

- [1] Chijun Sima*, **Yao Fu***, Man-Kit Sit, Liyi Guo, Xuri Gong, Feng Lin, Junyu Wu, Yongsheng Li, Haidong Rong, Pierre-Louis Aublin, and Luo Mai. Ekko: A large-scale deep learning recommender system with low-latency model update. *OSDI*, 2022.
- [2] Yao Fu, Leyang Xue, Yeqi Huang, Andrei-Octavian Brabete, Dmitrii Ustiugov, Yuvraj Patel, and Luo Mai. Serverlessllm: Locality-enhanced serverless inference for large language models. *arXiv preprint arXiv:2401.14351*, 2024.
- [3] Leyang Xue, Yao Fu, Zhan Lu, Luo Mai, and Mahesh Marina. Moe-infinity: Activation-aware expert offloading for efficient moe serving. *arXiv preprint arXiv:2401.14361*, 2024.
- [4] Jie Ren*, Xidong Feng*, Bo Liu*, Xuehai Pan*, **Yao Fu**, Luo Mai, and Yaodong Yang. Torchopt: An efficient library for differentiable optimization. *Journal of Machine Learning Research*, 24(367):1-14, 2023.
- [5] Yipeng Zhou, Xuezheng Liu, **Yao Fu**, Di Wu, Jessie Hui Wang, and Shui Yu. Optimizing the numbers of queries and replies in convex federated learning with differential privacy. *IEEE Transactions on Dependable and Secure Computing*, 2023.

Programming Languages

Advanced: C/C++, Python, Go

Intermediate: Java, Haskell, Matlab

Technical Skills

- Model serving libraries: Triton Inference Server, Ray Serve
- ML frameworks: PyTorch, HuggingFace ecosystem (Transformers, Accelerator, Safetensors)
- Containerization and orchestration: Docker, Kubernetes, Knative serving, KServe
- Network programming and API technologies: network sockets, gRPC, Flask, FastAPI
- System profiling & debugging tools: NVIDIA Nsight Systems, fio, iostat, perf

Awards

- Outstanding Graduates of Yat-sen Honors School, Sun Yat-sen University, 2021
- Outstanding Undergraduate Thesis Award, Sun Yat-sen University, 2021
- First Prize, Sun Yat-sen University Scholarship, 2020
- Parallel Fund Award, The 7th “Intel Cup” Parallel Application Challenge, 2019
- First Prize in Guangdong Province, China Undergraduate Mathematical Contest in Modeling, 2019
- Second Prize, Sun Yat-sen University Scholarship, 2018-2019

Talks

16th USENIX Symposium on Operating Systems Design and Implementation (OSDI)

Jul. 2022

Sixth Annual UK System Research Challenges Workshop

Dec. 2021

Teaching and Academic Services

- HPCA 2024, Artifact Evaluation Committee
- Extreme Computing 2023 Fall/2021 Fall, Marker